Chengpeng Wang

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RESEARCH INTEREST

Software Engineering, Programming Languages, and Machine Learning, with a focus on developing program analysis techniques that enhance the reliability and performance of modern software systems, with particular interests in AI-assisted software development and AI agents.

EDUCATION

Hong Kong University of Science and Technology *Aug 2019 - Dec 2023*
Ph.D. in Computer Science and Engineering

Tsinghua University *Aug 2016 - Jul 2019*
M.S. in Software Engineering

Tsinghua University *Aug 2012 - Jul 2016*
B.S. in Software Engineering, Minor in Math

ACADEMIC AND INDUSTRY APPOINTMENTS

Assistant Professor, National University of Singapore, Singapore *Jun 2026 - Now*
School of Computing

Postdoctoral Fellow, Purdue University, West Lafayette, IN, USA *Mar 2024 - May 2026*
Neuro-symbolic code auditing, working with Prof. Xiangyu Zhang

Research Intern at Veridise, Remote *May 2023 - Jul 2023*
Static bug detection of zero-knowledge proof circuits

Research Intern at Ant Group, Shenzhen, China *Feb 2021 - Aug 2022*
Performance optimization of database-backed application

Research Intern at Sourcebrella, Shenzhen, China *Feb 2019 - Aug 2019*
Inconsistency detection of build systems

HONORS AND AWARDS

ISSTA Distinguished Reviewer Award, ACM SIGSOFT *2026*
COLM Oral Spotlight Paper, COLM *2026*
ASE Distinguished Reviewer Award, ACM SIGSOFT *2025*
NeurIPS Spotlight Paper, NeurIPS *2025*
Postdoc Travel Award, Purdue University *2024*
ACM SIGARCH Best Paper Award, ASPLOS *2024*
UGC Research Travel Grant, HKUST *2022, 2023*
Ant Group Outstanding Collaboration Award *2023*
ACM SIGPLAN Distinguished Paper Award, OOPSLA *2022*

PUBLICATIONS

PREPRINTS

[P1] Zhiwei Lin, Jiahao Liu, Yan Liu, and **Chengpeng Wang**, To Cache or Not to Cache: Understanding and Mitigating Safety Incompatibility in Cross-Model KV Cache Reuse. **arXiv:2610.15721**, Oct 11, 2026.

[P2] Yuan Li, Hanyun Jiang, Guowei Tian, **Chengpeng Wang**, and Peisen Yao, DatalogBench: Evaluating Large Language Models on Text-to-Datalog Synthesis. **arXiv:2609.37233**, Sep 29, 2026.

[P3] Haomin Qi, Xiangzhe Xu, Yiming Huang, Jingbo Shang, and **Chengpeng Wang**, WitnessGym: Benchmarking Coding Agents on the Construction of Bug Witnesses. **arXiv:2609.36635**, Sep 29, 2026.

[P4] Xuan Chen, **Chengpeng Wang**, Lu Yan, and Xiangyu Zhang, SLBench: Evaluating How LLM Agents Follow Logical Relations in Skills. **arXiv:2607.09016**, Jul 10, 2026.

JOURNAL PUBLICATIONS

[J1] **Chengpeng Wang**, Zhuo Zhang, Xiangzhe Xu, Mingwei Zheng, Guannan Wei, and Xiangyu Zhang, Neurosymbolic Code Auditing, In **FTPL 2026**: Foundations and Trends in Programming Languages.

[J2] Chao Wang, Li Lin, **Chengpeng Wang**, Jiafeng Huang, Congxia Wu, and Rongxin Wu, ReachCheck: Compositional Library-Aware Call Graph Reachability Analysis in the IDEs, In **TOSEM 2025**: Transactions on Software Engineering and Methodology

[J3] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Operation-Level Memory Sanitization with Segment Folding, In **TOCS 2024**: The ACM Transactions on Computer Systems. (Invited extension of [C19])

[J4] Rongxin Wu, Zhiling Huang, Zige Tian, **Chengpeng Wang**, and Xiangyu Zhang, PackHunter: Recovering Missing Packages for C/C++ Projects, In **TSE 2024**: Transactions on Software Engineering

[J5] Yi Sun, **Chengpeng Wang**, Gang Fan, Qingkai Shi, and Xiangyu Zhang, Fast and Precise Static Null Exception Analysis with Synergistic Preprocessing, In **TSE 2024**: Transactions on Software Engineering

[J6] Wensheng Tang, Dejun Dong, Shijie Li, **Chengpeng Wang**, Peisen Yao, Jinguo Zhou, and Charles Zhang, Octopus: Scaling Value-Flow Analysis via Parallel Collection of Realizable Path Conditions, In **TOSEM 2023**: Transactions on Software Engineering and Methodology

[J7] **Chengpeng Wang**, Wenyang Wang, Peisen Yao, Qingkai Shi, Jinguo Zhou, Xiao Xiao, and and Charles Zhang, Anchor: Fast and Precise Value-Flow Analysis for Containers via Memory Orientation, In **TOSEM 2022**: Transactions on Software Engineering and Methodology

CONFERENCE PUBLICATIONS

[C1] Zhou Xuan, Xiangzhe Xu, Mingwei Zheng, Louis Zheng-Hua Tan, Jinyao Guo, Tiantai Zhang, Le Yu, **Chengpeng Wang**, and Xiangyu Zhang, TTPScan: An LLM Agent for TTP Recognition in Malware Binaries, In **ICSE 2027**: International Conference on Software Engineering.

[C2] Jinyao Guo, **Chengpeng Wang**, Dominic Deluca, Jinjie Liu, Zhuo Zhang, and Xiangyu Zhang, BugScope: Learn to Find Bugs Like Human, In **COLM 2026**: Conference on Language Modeling. 🏆 Oral Spotlight

[C3] Jinyao Guo, Yaoyang Ye, Zhihao Guo, Mingwei Zheng, Tiantai Zhang, Xuwei Liu, Xiaolong Jin, Congyu Liu, Xiangzhe Xu, Zian Su, Ziyun Song, Zhou Xuan, Xiao Cheng, **Chengpeng Wang**, and Xiangyu Zhang, Benchmarking Explainable Vulnerability Detection with Calling Contexts, In Findings of **EMNLP 2026**: Empirical Methods in Natural Language Processing.

[C4] Yifei Gao, **Chengpeng Wang**, Pengxiang Huang, Xuwei Liu, Mingwei Zheng, and Xiangyu Zhang, Raw Pointer Rewriting with LLMs for Translating C to Safer Rust, In Findings of **ACL 2026**: Findings of the Association for Computational Linguistics.

[C5] Hanyun Jiang, Peisen Yao, Kaiyue Li, Tingting Lin, **Chengpeng Wang**, and Kui Ren, HintPilot: LLM-based Compiler Hint Synthesis for Code Optimization, In Findings of **ACL 2026**: Findings of the Association for Computational Linguistics.

[C6] **Chengpeng Wang**, Yifei Gao, Wuqi Zhang, Xuwei Liu, Jinyao Guo, Mingwei Zheng, Qingkai Shi, and Xiangyu Zhang, NESA: Relational Neuro-Symbolic Static Program Analysis, In **FSE 2026**: International Conference on the Foundations of Software Engineering.

[C7] Ming Liang, Xiaoheng Xie, Gehao Zhang, Xunjin Zheng, Wei Jiang, **Chengpeng Wang**, Gang Fan, and Peng Di, RepoFuse: A Dual-Context Approach to Repository-Level Code Completion at Industrial Scale, In **FSE 2026**, Industry Track: International Conference on the Foundations of Software Engineering.

[C8] Mingwei Zheng, **Chengpeng Wang**, Xuwei Liu, Jinyao Guo, Shiwei Feng, and Xiangyu Zhang, RFCAudit: AI Agent for Auditing Protocol Implementations Against RFC Specifications, In **ASE 2025**: International Conference on Automated Software Engineering.

[C9] Danning Xie, Mingwei Zheng, Xuwei Liu, Jiannan Wang, **Chengpeng Wang**, Lin Tan, and Xiangyu Zhang, CORE: Benchmarking LLMs' Code Reasoning Capabilities through Static Analysis Tasks, In **NeurIPS 2025**: Annual Conference on Neural Information Processing Systems. 🏆 Spotlight Paper

[C10] Jinyao Guo[#], **Chengpeng Wang**[#], Xiangzhe Xu, Zian Su, and Xiangyu Zhang. RepoAudit: An Autonomous LLM-Agent for Repository-Level Code Auditing. In **ICML 2025**: International Conference on Machine Learning.

[C11] Yuan Li, Peisen Yao, Kan Yu, **Chengpeng Wang**, Yaoyang Ye, Song Li, Meng Luo, Yepang Liu, and Kui Ren. Understanding Industry Perspectives of Static Application Security Testing Evaluation, In **FSE 2025**: International Conference on the Foundations of Software Engineering.

[C12] Mingwei Zheng, Danning Xie, Qingkai Shi, **Chengpeng Wang**, and Xiangyu Zhang, Validating Network Protocol Parsers with Traceable RFC Document Interpretation, In **ISSTA 2025**: International Symposium on Software Testing and Analysis.

[C13] Wei Chen, Bowen Zhang, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, Seal: Towards Diverse Specification Inference for Linux Interfaces from Security Patches, In **EuroSys 2025**: European Conference on Computer Systems.

[C14] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, Xiaoheng Xie, and Xiangyu Zhang, LLMDFA: Analyzing Dataflow in Code with Large Language Models, In **NeurIPS 2024**: Annual Conference on Neural Information Processing Systems

[C15] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, and Xiangyu Zhang, Sanitizing Large Language Models in Bug Detection with Data-Flow, In Findings of **EMNLP 2024**: Empirical Methods in Natural Language Processing

[C16] Li Lin, Zongyin Hao, **Chengpeng Wang**, Zhuangda Wang, Rongxin Wu, and Gang Fan, SQLess: Dialect-Agnostic SQL Query Simplification, In **ISSTA 2024**: International Symposium on Software Testing and Analysis

[C17] **Chengpeng Wang**, Jipeng Zhang, Rongxin Wu, and Charles Zhang, DAInfer: Inferring API Aliasing Specifications from Library Documentation via Neurosymbolic Optimization, In **FSE 2024**: International Conference on the Foundations of Software Engineering

[C18] Bowen Zhang, Wei Chen, Peisen Yao, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, SIRO: Empowering Version Compatibility in Intermediate Representations via Program Synthesis, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems

[C19] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Memory Sanitization with Segment Folding, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems 🏆 ACM SIGARCH Best Paper Award

[C20] Rongxin Wu, Yuxuan He, Jiafeng Huang, **Chengpeng Wang**, Wensheng Tang, Qingkai Shi, Xiao Xiao, and Charles Zhang, LibAlchemy: A Two-Layer Persistent Summary Design for Taming Third-Party Libraries in Static Bug-Finding Systems, In **ICSE 2024**: International Conference on Software Engineering

[C21] Wensheng Tang[#], **Chengpeng Wang**[#], Peisen Yao, Rongxin Wu, Xianjin Fu, Gang Fan, and Charles Zhang, DCLink: Bridging Data Constraint Changes and Implementations in FinTech Systems, In **ASE 2023** : International Conference on Automated Software Engineering

[C22] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Gang Fan, Charles Zhang, Synthesizing Conjunctive Queries for Code Search, In **ECOOP 2023**: European Conference on Object-Oriented Programming

[C23] Zongyin Hao, Quanfeng Huang, **Chengpeng Wang**, Jianfeng Wang, Yushan Zhang, Rongxin Wu, and Charles Zhang, Pinolo: Detecting Logical Bugs in Database Management Systems with Approximate Query Synthesis, In **ATC 2023**: USENIX Annual Technical Conference

[C24] **Chengpeng Wang**, Gang Fan, Peisen Yao, Fuxiong Pan, and Charles Zhang, Verifying Data Constraint Equivalence in FinTech Systems, In **ICSE 2023**: International Conference on Software Engineering

[C25] Rongxin Wu, Minglei Chen, **Chengpeng Wang**, Gang Fan, Jiguang Qiu, and Charles Zhang, Accelerating Build Dependency Error Detection via Virtual Build, In **ASE 2022** : International Conference on Automated Software Engineering

[C26] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Qingkai Shi, and Charles Zhang: Complexity-Guided Container Replacement Synthesis, In **OOPSLA 2022**: SIGPLAN Conference on Objected Oriented Programming, Systems, Languages and Applications 🏆 ACM SIGPLAN Distinguished Paper Award

[C27] Gang Fan, **Chengpeng Wang**, Rongxin Wu, Xiao Xiao, Qingkai Shi, and Charles Zhang: Escaping Dependency Hell: Finding Build Dependency Errors with the Unified Dependency Graph, In **ISSTA 2020**: International Symposium on Software Testing and Analysis

WORKSHOP and POSTER PUBLICATIONS

[W1] Ming Liang, Xiaoheng Xie, Gehao Zhang, Xunjin Zheng, Peng Di, Wei Jiang, Hongwei Chen, **Chengpeng Wang**, and Gang Fan, RepoGenix: Dual Context-Aided Repository-Level Code Completion with Language Models, In Poster Track, **ASE 2024**: International Conference on Automated Software Engineering

[W2] **Chengpeng Wang**, CodeSpider: Automatic Code Querying with Multi-modal Conjunctive Query Synthesis, In SRC track, **SPLASH 2022**: SIGPLAN conference on Systems, Programming, Languages, and Applications: Software for Humanity, Student Research Competition

[W3] **Chengpeng Wang**, Yixiao Yang, Han Liu, and Le Kang: Statistical API Completion Based on Code Relevance Mining, In **MAINT 2019**: International Workshop on Mining and Analyzing Interaction Histories

ACADEMIC SERVICES

Organizing Committee Member:

International Conference on the Foundations of Software Engineering (FSE), Student Research Competition (SRC) *2027*
International Workshop on Language Models and Programming Languages (LMPL) *2025*

Program Committee Member:

ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA) *2027*
International Workshop on Language Models and Programming Languages (LMPL) *2026*
International Workshop on Principles of Agentic Engineering (PAgE) *2026*
International Conference on the Foundations of Software Engineering (FSE) *2025, 2026, 2027*
International Conference on Software Engineering (ICSE) *2027*
Annual Conference on Neural Information Processing Systems (NeurIPS) *2025, 2026*
ACL Rolling Review (ARR) *2026*
ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI) *2026*
International Symposium on Software Testing and Analysis (ISSTA) *2025, 2026*
International Workshop on Large Language Models for Code (LLM4Code) *2026*
International Conference on Automated Software Engineering (ASE) *2025*
International Conference on AI Foundation Models and Software Engineering (Forge) *2024, 2025*
SPLASH, Student Research Contest Track *2024*
International Conference on Automated Software Engineering (ASE), Industrial Track *2024*
International Symposium on Software Reliability Engineering (ISSRE) *2024, 2025*

Journal Reviewer:

Transactions on Software Engineering and Methodology (TOSEM) 2025
Transactions on Software Engineering (TSE) 2024
Communications of the ACM (CACM) 2026

Artifact Evaluation Committee Member:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA) *2024*
ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI) *2023*
International Conference on the Foundations of Software Engineering (FSE) *2022*
International Symposium on Software Testing and Analysis (ISSTA) *2022*

Volunteer:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA) *2022*
Student Volunteer at International Symposium on Software Testing and Analysis (ISSTA) *2019*
Long-term mentor of SIGPLAN-M *2024, 2025, 2026*

TEACHING

CS 6215: Advanced Topics in Program Analysis, NUS	*Fall 2026*
CS 510: Software Engineering, Purdue University	*Fall 2025*
CS 592: AI and Security, Purdue University	*Fall 2024*
COMP 3021: Java Programming, HKUST	*Spring/Fall 2022/2023*
COMP 4631: Computer and Communication Security, HKUST	*Fall 2021*
COMP 3111/H: Software Engineering, HKUST	*Fall 2020*
COMP 2011: Programming with C++, HKUST	*Spring 2020*
Haskell: Functional Language Programming, THU	*Spring 2019*
Automaton and Formal Logic, THU	*Fall 2018*

PRESENTATIONS AND INVITED TALKS

Neuro-Symbolic Code Auditing for Software Reliability in the AI Era
National University of Singapore, Singapore Management University — Dec 2025

Static Code Auditing for Software Quality Assurance in the AI Era
Tufts University, Boston — Oct 2025

RepoAudit: Human-like AI Auditor for Code Repositories
RSA Conference 2025, San Francisco — Apr 2025

Advances in AI-powered Code Security: Next-Level Bug Detection
Uber's Programming Systems Team. Virtual — Feb 2025
CodeQL@GitHub. Virtual — Jan 2025

Neuro-Symbolic Static Analysis for Reliable Software Systems
University of British Columbia, Columbia University — Aug 2025
UC Davis, UIUC, UNSW — May 2025
AST Lab, ETH Zurich. Virtual — Mar 2025
Department of Computing, Hong Kong Polytechnic University — Jan 2025
Institute of Data Science, The University of Hong Kong — Dec 2024

Neuro-Symbolic Static Analysis for Reliable and Performant Software Systems
School of Computer Science, Nanjing University. Nanjing, China — Dec 2024

When Static Analysis Meets Large Language Models: A Neuro-Symbolic Approach
Ant Group. Virtual — Dec 2024
Boston Computation Club. Virtual — Nov 2024
AI-CyberSecurity Research Lunch Talk, Texas A&M University (TAMU). Virtual — Oct 2024
School of Software, Tsinghua University. Virtual — Oct 2024

Towards Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis
School of Informatics, Xiamen University. Xiamen, China — Aug 2023

Synthesizing Conjunctive Queries for Code Search
ByteDance. Virtual — Jun 2023

Complexity-Guided Container Replacement Synthesis
AST Lab, ETH Zurich. Virtual — Mar 2023